

# **CLOUD-NATIVE MICROSERVICES E-COMMERCE PLATFORM WITH MONITORING & SECURITY**

**MINI PROJECT**

**22CB907 CLOUD MICROSERVICES AND APPLICATIONS**

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# PROBLEM STATEMENT

In modern urban environments, traditional e-commerce systems are often built using monolithic architectures, which makes them difficult to scale, maintain, and secure. As user demand increases, these systems face performance bottlenecks, limited flexibility, and higher downtime risks. Additionally, lack of proper monitoring and security mechanisms leads to vulnerabilities, data breaches, and inefficient resource utilization. There is a need for a scalable, secure, and efficient cloud-based solution that can handle dynamic workloads while ensuring reliability, modularity, and real-time monitoring.



# ABSTRACT

This project presents a Cloud-Native Microservices E-Commerce Platform designed using containerization and modern DevOps practices. The system is divided into independent microservices such as user-service, product-service, order-service, and an API gateway, enabling scalability and flexibility. Docker is used for containerization, and Docker Compose is used to orchestrate multiple services efficiently. The platform ensures better resource utilization, security, and monitoring capabilities. By adopting microservices architecture and cloud-native principles, the system provides improved performance, fault isolation, and easy deployment, making it suitable for real-world scalable applications.

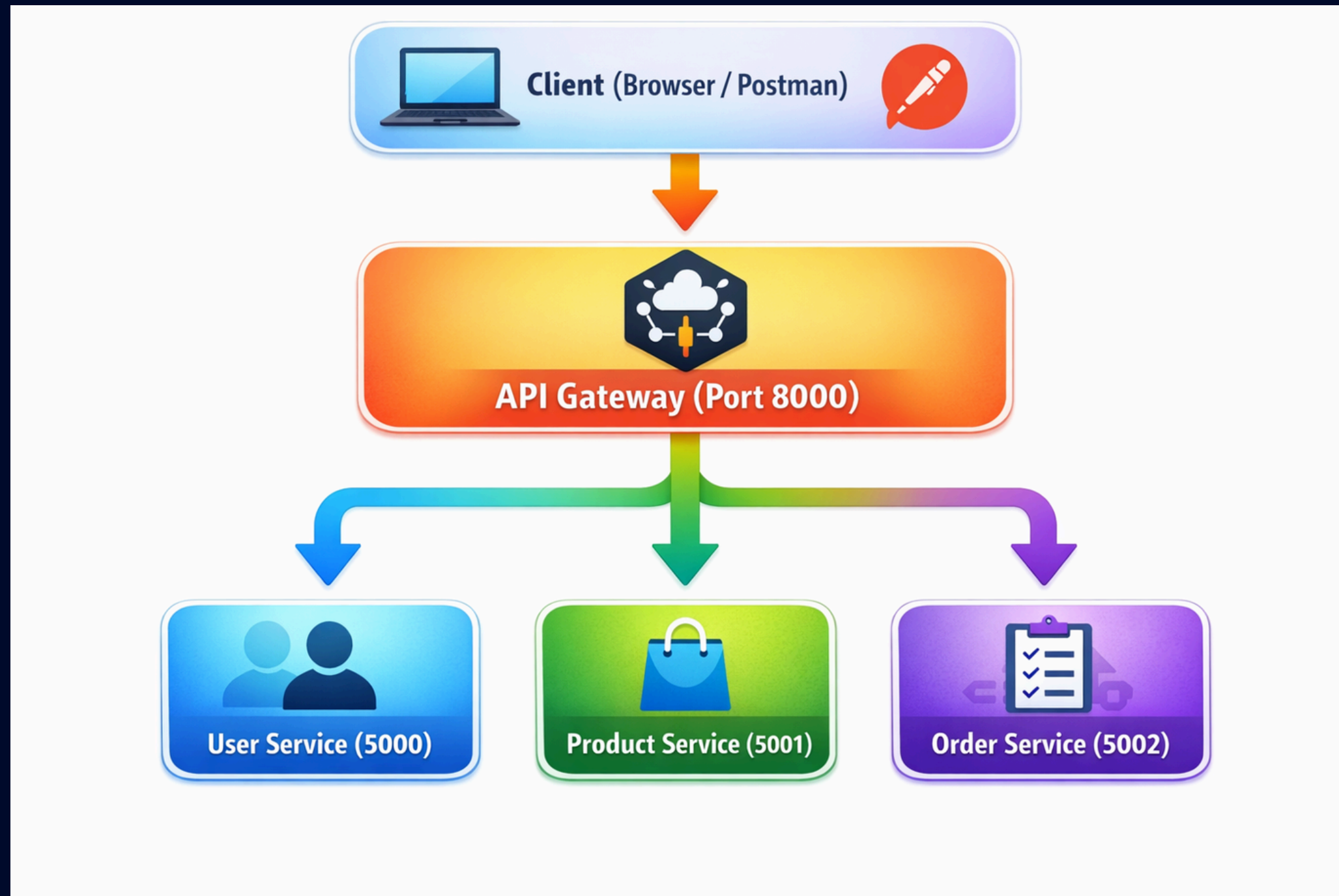


# BRIEF DESCRIPTION

This project is a microservices-based e-commerce backend system where:

- Each service works independently
- API Gateway handles all requests
- Docker manages containerization
- System is scalable and cloud-ready

# SYSTEM ARCHITECTURE



# FLOW CHART





# TECHNOLOGIES USED

- Python (Flask)
- Docker
- Docker Compose
- Microservices Architecture
- REST API
- Cloud Computing Concepts

# PROJECT STRUCTURE





# SECURITY FEATURES

- Service isolation using containers
- API-based communication
- Independent service control

# ADVANTAGES

- Scalable
- Easy to maintain
- Fault isolation
- Faster deployment
- Cloud-ready

# MONITORING

- Container-based monitoring possible
- Logs from each service
- Scalable cloud deployment



# CONCLUSION

The project successfully demonstrates how cloud-native microservices architecture can be used to build scalable and efficient applications. By using Docker and modular services, the system ensures flexibility, reliability, and improved performance



➔ **GITHUB LINK:**

<https://github.com/manju-2005/Cloud-Native-Microservices-E-Commerce-Platform-with-Monitoring-Security>:

**THANK YOU**